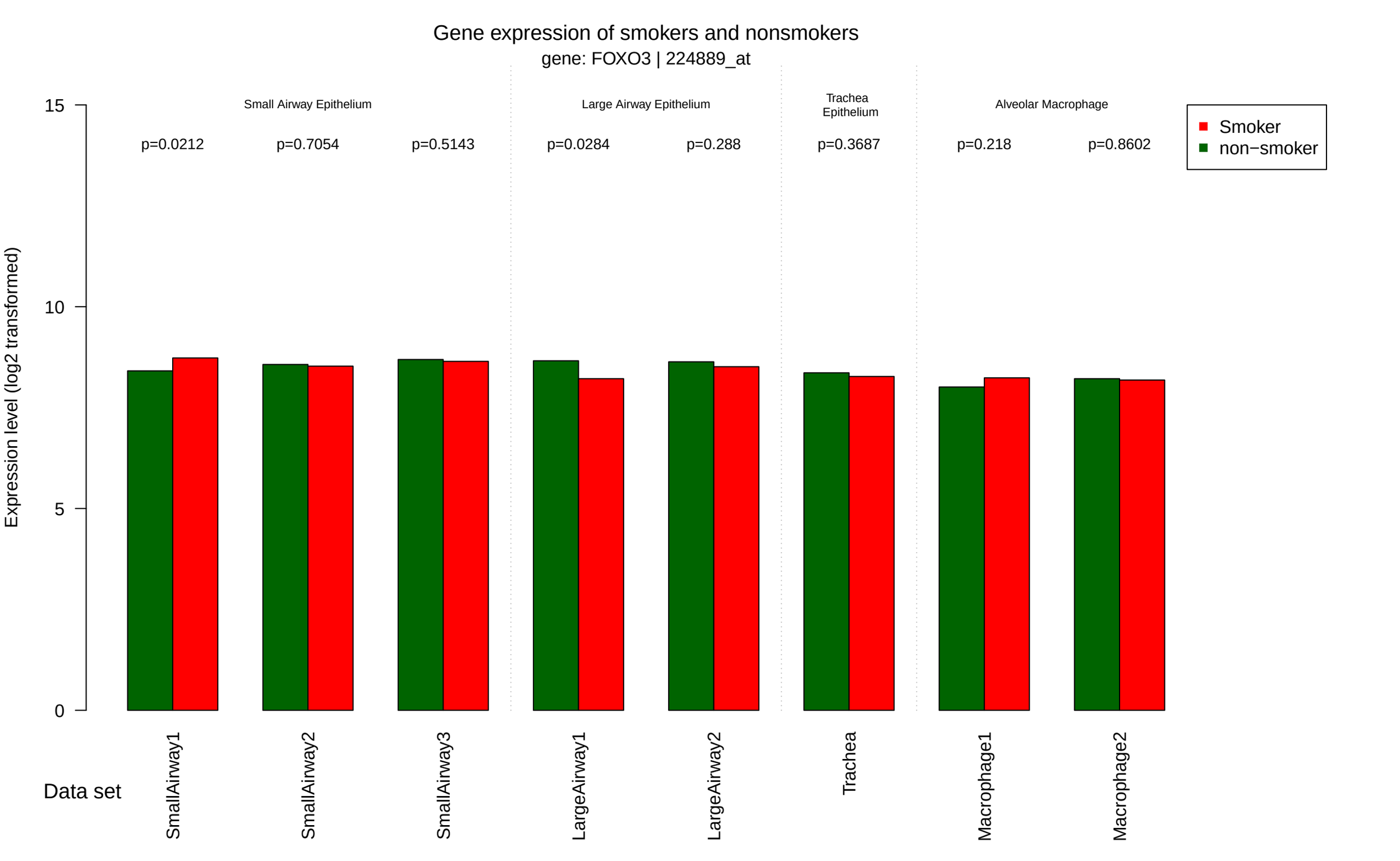
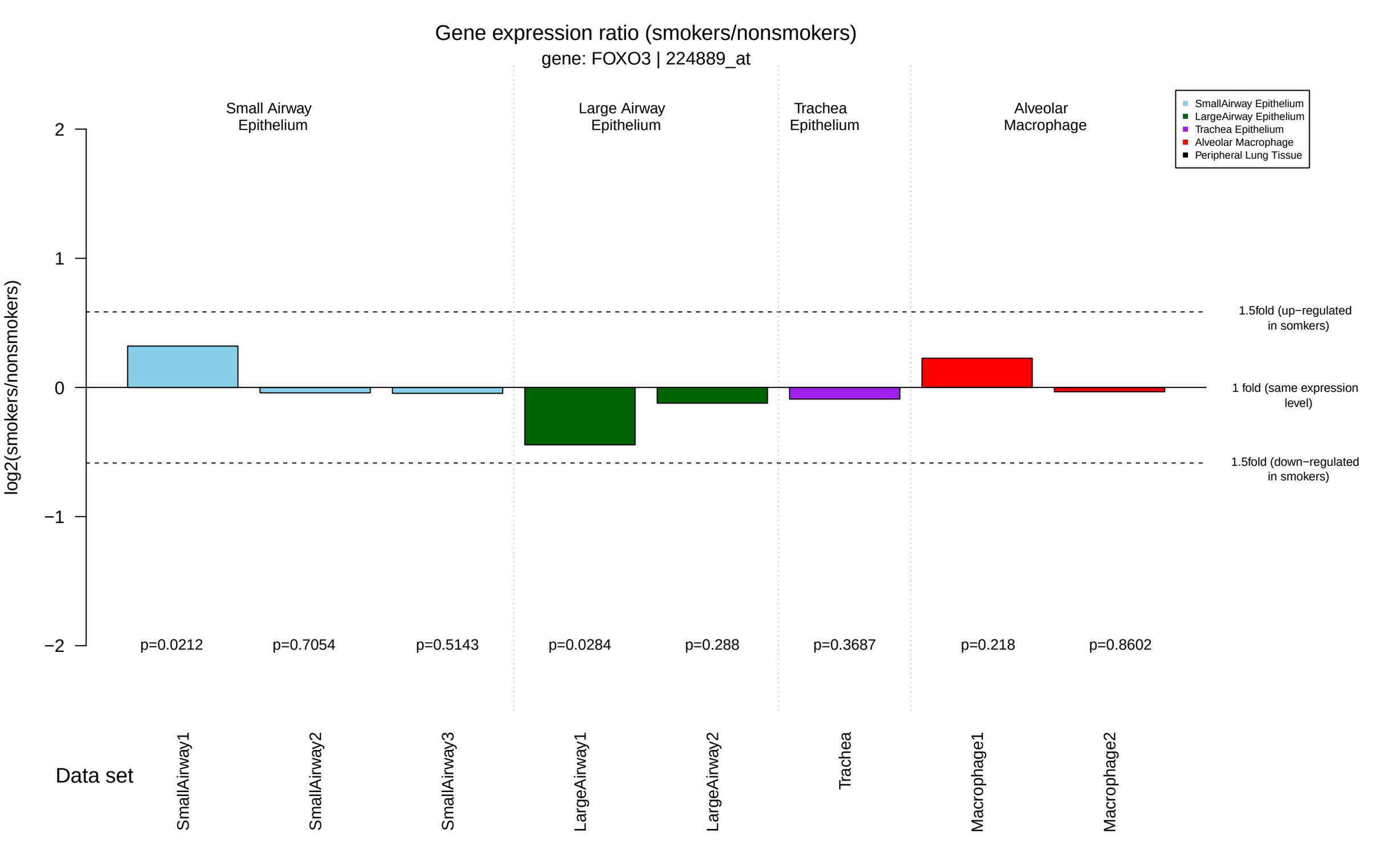
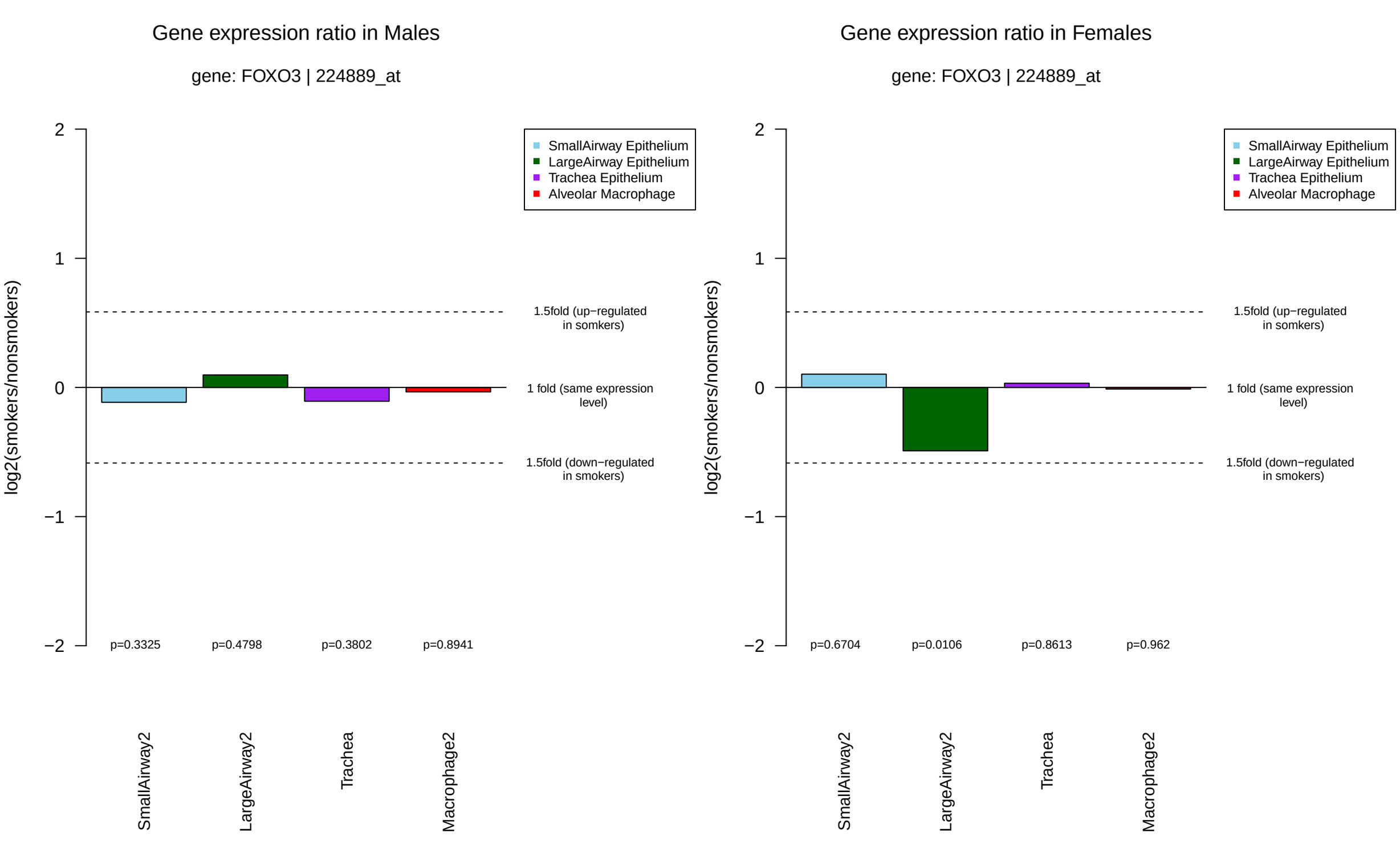
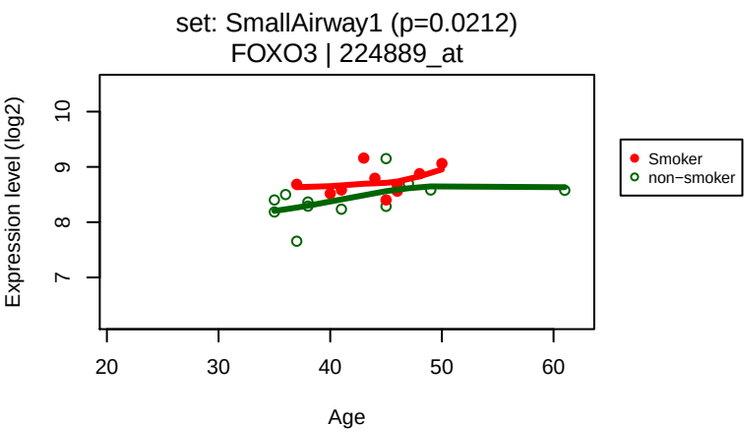
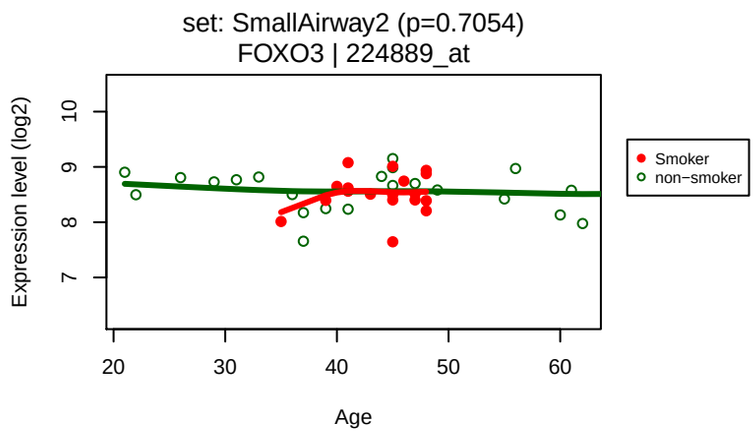
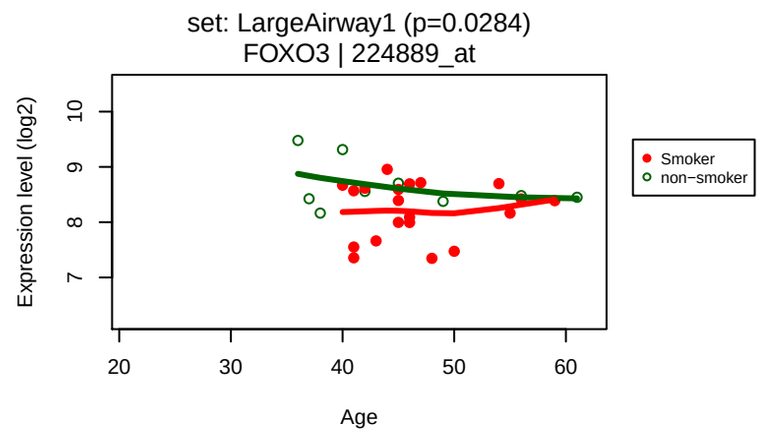

Gene expression of smokers and nonsmokers
gene: FOXO3 | 224889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0212
p=0.7054
p=0.5143
p=0.0284
p=0.288
p=0.3687
p=0.218
p=0.8602
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

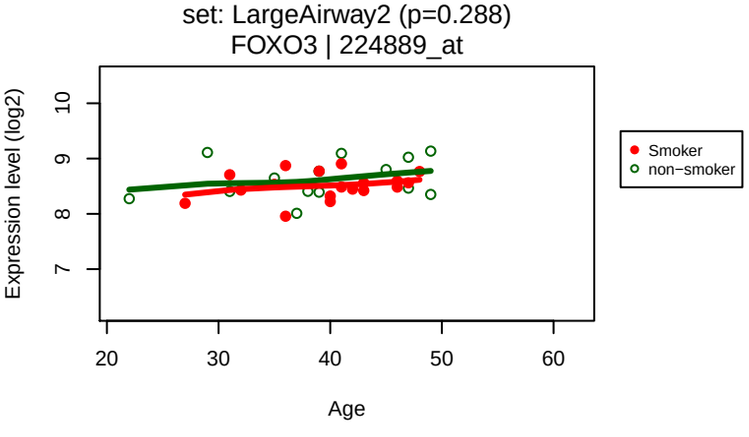

Gene expression ratio (smokers/nonsmokers)
gene: FOXO3 | 224889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0212
p=0.7054
p=0.5143
p=0.0284
p=0.288
p=0.3687
p=0.218
p=0.8602
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

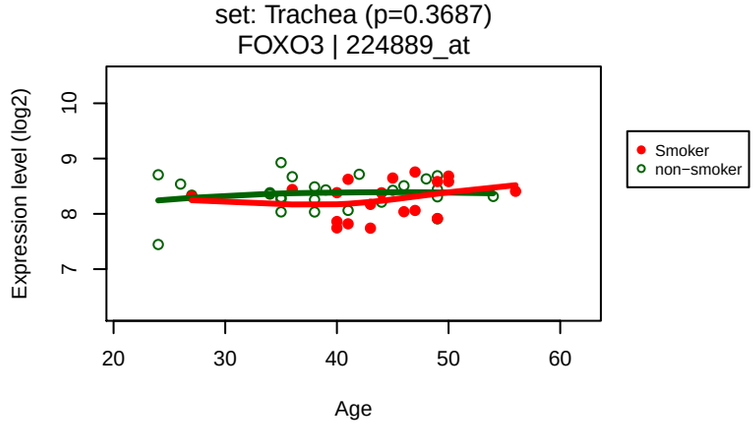

Gene expression ratio in Males
gene: FOXO3 | 224889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3325
p=0.4798
p=0.3802
p=0.8941
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FOXO3 | 224889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6704
p=0.0106
p=0.8613
p=0.962
SmallAirway2
LargeAirway2
Trachea
Macrophage2

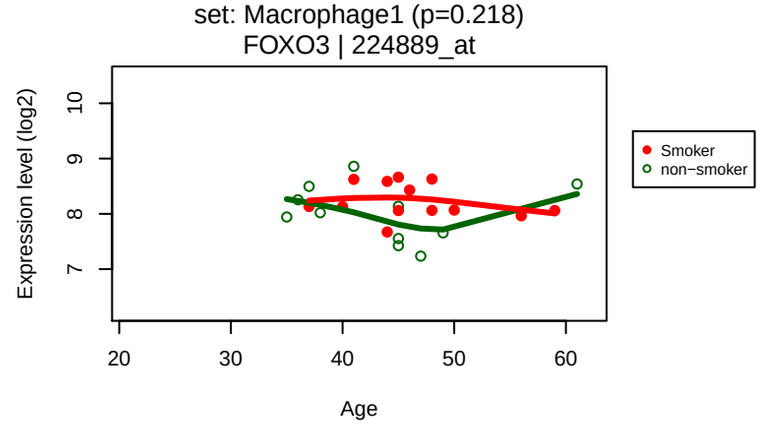

set: SmallAirway1 (p=0.0212)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker

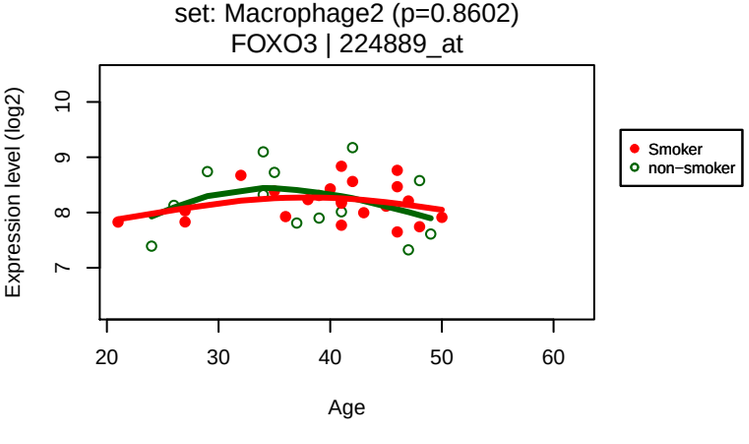

set: SmallAirway2 (p=0.7054)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker

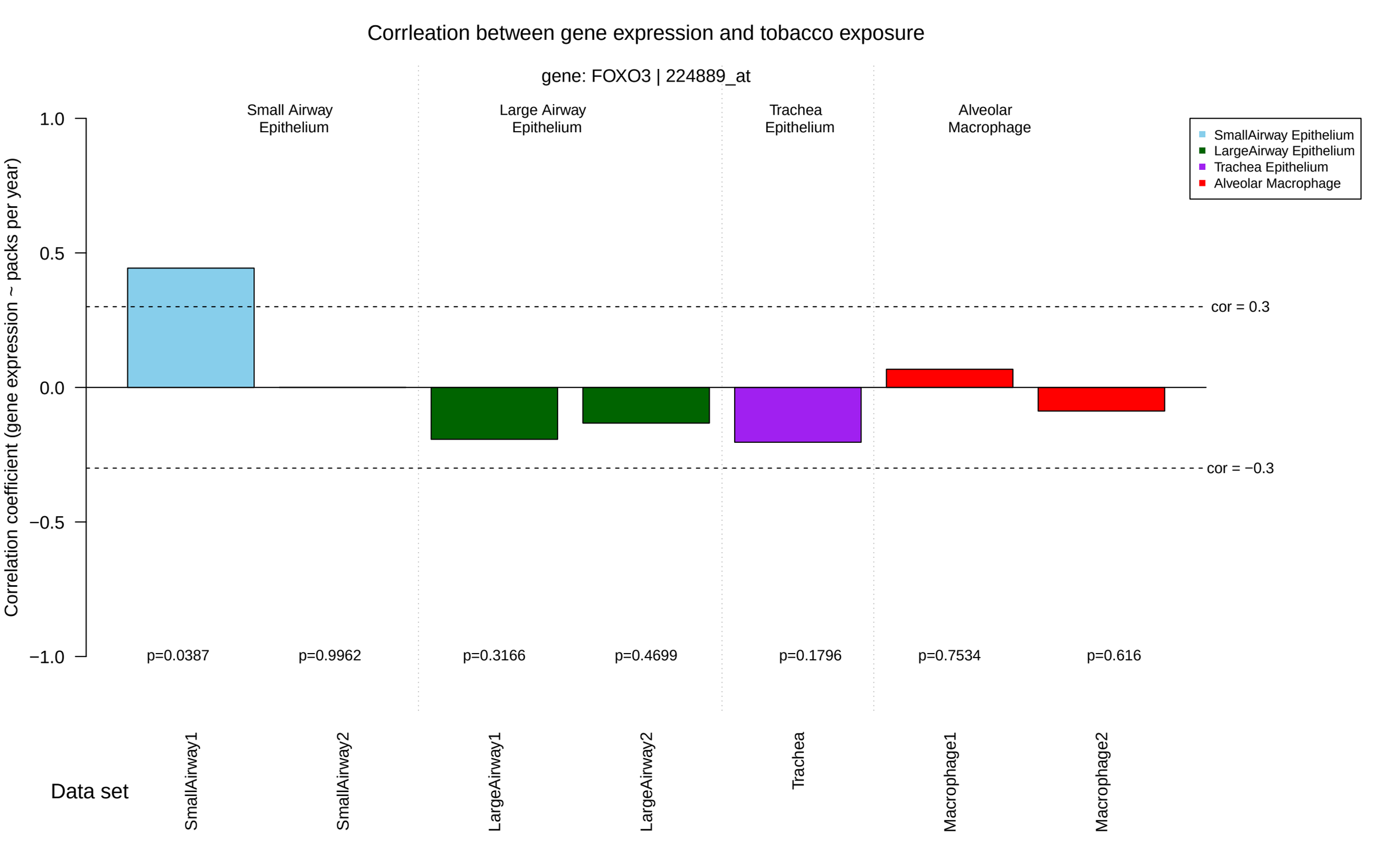

set: LargeAirway1 (p=0.0284)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.288)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3687)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.218)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8602)
FOXO3 | 224889_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FOXO3 | 224889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0387
p=0.9962
p=0.3166
p=0.4699
p=0.1796
p=0.7534
p=0.616
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2